



EXAM PAPERS PRACTICE

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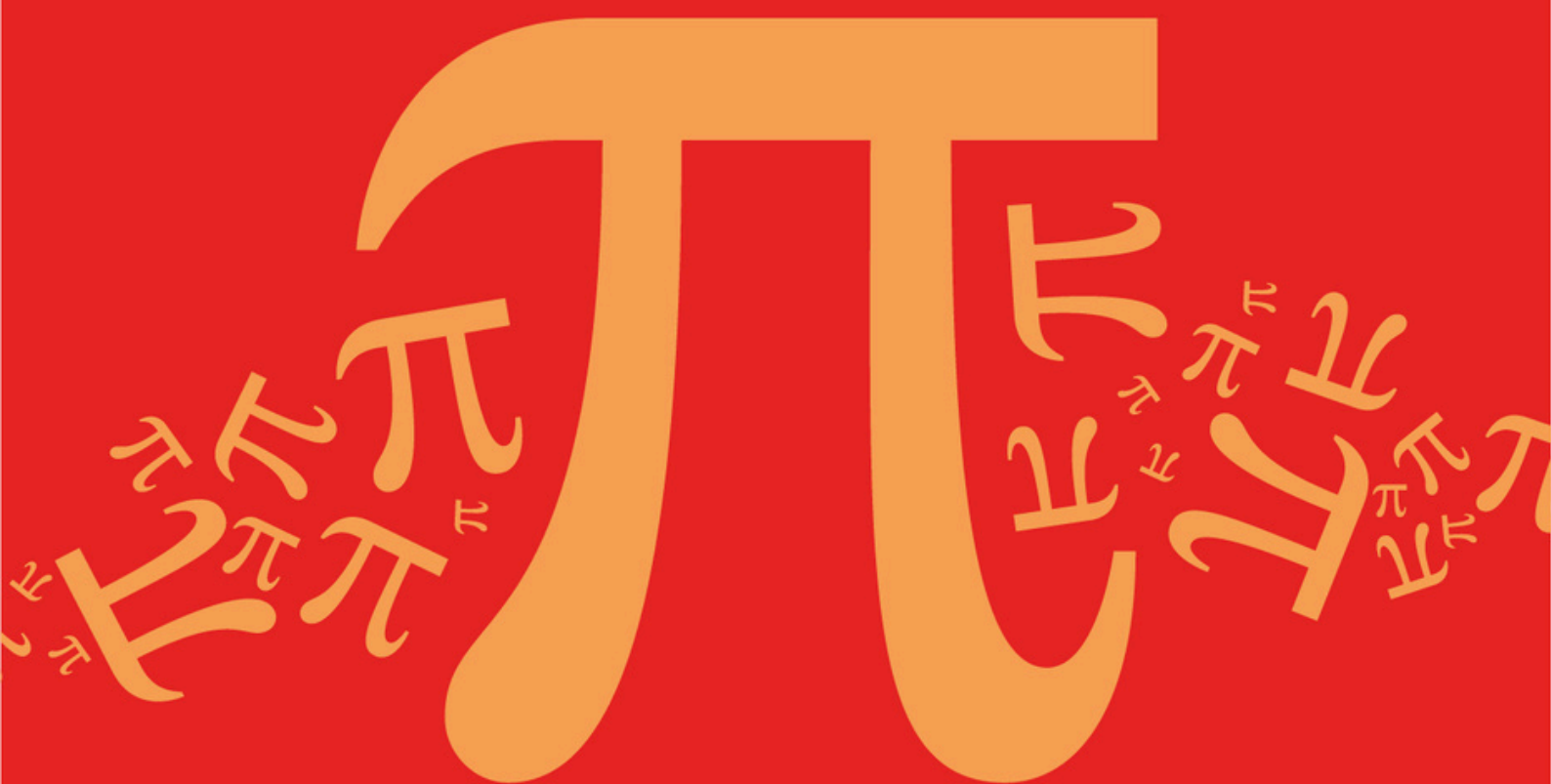
Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

DB: Networks

Sub topic

DB5: Evaluate, Modify, and Refine Models

Mark Scheme

AQA A Level Further Mathematics (7367)

Mark Scheme

Topic	DB: Networks
Sub topic	DB5: Evaluate, Modify, and Refine Models

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

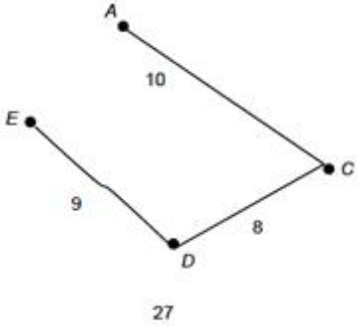
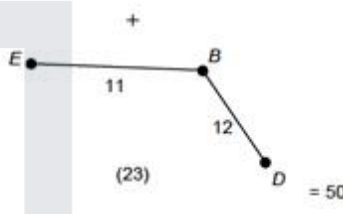
Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Gives a plausible reason, in the context of the question, as to what *** could mean, such as no cycle path exists	AO2.4	E1	A cable cannot be laid between these two districts as there may be a river or housing estate in the way
(b)	Identifies the problem as a route inspection problem by noting that <i>B</i> , <i>G</i> , <i>H</i> and <i>O</i> are odd-degree nodes. (PI)	AO3.1b	M1	odd nodes: <i>B</i> , <i>G</i> , <i>H</i> and <i>O</i>
	Finds the shortest distance between each pair of odd nodes (at least four correct) OR determines correctly values for the *** entries in the table, with clear reasoning.	AO1.1b	M1	<i>B-G</i> : 2.5; <i>B-H</i> : 5.6; <i>B-O</i> : 4.3; <i>G-H</i> : 3.1; <i>G-O</i> : 6.8; <i>H-O</i> : 6.7. OR <i>B-H</i> : 5.6 (via <i>G</i>) and <i>G-O</i> : 6.8 (via <i>B</i>)
	Determines correctly the minimum total distance the engineer will cover during the journey. CAO	AO1.1a	A1	Minimum pair of shortest distances is <i>B-O</i> and <i>G-H</i> , with a minimum extra distance of 7.4 miles $37.2 + 7.4 = 44.6$ miles
	Determines the minimum time in minutes that the engineer could complete the journey in by using 'their' minimum total distance and the average speed of 12 miles per hour.	AO3.2a	A1F	$(44.6 / 12) \times 60 = 223$ minutes.
Total 5 marks				

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Finds the correct Hamiltonian cycle starting	AO1.1a	M1	<i>A C D E B A</i>

	at A using the nearest neighbour algorithm			(10 + 8 + 9 + 11 + 15)
	Correctly determines and states the upper bound	AO1.1b	A1	= 53
(b)	Correctly determines a minimum spanning tree connecting A, D, E and C (or 27 seen)	AO1.1b	B1	
	Correctly identifies the two shortest arcs connected to B	AO1.1a	M1	
	Calculates the sum of 'their' minimum spanning tree and the two shortest arcs	AO1.1b	A1	
(c)	Infers correctly that the optimal Hamiltonian cycle has a weight between 50 and 53 minutes, and that there is currently no further information to suggest a route of less than 53 minutes exists as the lower bound is not a Hamiltonian cycle	AO2.2b	R1	The optimal time for a tour lies between 50 and 53 minutes, and the lower bound is not a Hamiltonian cycle, so we cannot be sure that a Hamiltonian cycle of less than 53 minutes exists.
(d)	Evaluates answers in parts (a) and (b) with reference to a plausible reason why Charlotte has taken longer than the typical tourist	AO3.5a	E1	The model only applies to a typical tourist. Charlotte takes longer than the typical tourist because she walks more slowly than the typical tourist.
(e)	Gives a plausible reason for why the times taken to walk between the monuments would likely be different during the evening	AO3.5b	E1	It is likely there will be less traffic at night, meaning that tourists travel time between each monument is likely to be less than the times during the morning. For example, they would not have to wait as long to walk across roads. OR At night there may be an increase in the number of pedestrians and it will therefore take longer to walk between the

			monuments.
Total 8 marks			

Q3.

(a)

$$\begin{array}{rcl}
 BD+FH & = & \left[\begin{array}{c} 210+210 \\ 200+180 \\ 260+340 \end{array} \right] = 420 \\
 BF+DH & = & \\
 BH+DF & = &
 \end{array}$$

These 3 sets of pairs

3 correct totals, 2 correct totals

(MIN) = 2430 + 380

2430 + their smallest of three pair totals

= 2810

CSO

M1

A2,1

m1

A1

5

(b) 2430 + 340 (DF) = 2770

2430 + their DF

B1F

1

(c) (i) 2430 + 180 (DH) = 2610

2430 + their min (must have scored M1)

B1F

1

(ii) B, F only

B1

1

[8]

Q4.

(a) (i)

EB	$\begin{pmatrix} 5 \end{pmatrix}$
EH	7
AB	8
HI	9
AD	10
DG	4
EF	12
FC	$\begin{pmatrix} 6 \end{pmatrix}$

Prim's, MST, 6+ edges (no cycles), edges not lengths or vertices, with first 2 edges correct

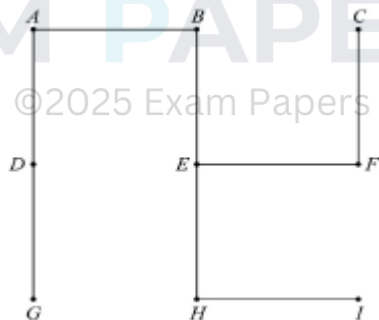
8 edges

AB 3rd

All correct

(ii) 61

(iii)



6+ edges, connected, no cycles

Correct, including labelling

- (b) Delete BA , BE and reconnect with 1 edge
 or
 a spanning tree with 7 edges not including B
 (either as a list or diagram)
P1 from their diagram in (iii)

M1

B1

A1

A1

4

B1

1

M1

A1

2

M1

$$(61 - 13 + 11) = 59$$

Note: 59 scores 2/2

A1

2

[9]

Q5.

(a) (i)

AC
 CH
 FH
 CE
 CD
 GH
 DB

(or ED)

Prim's, ST, 5+ edges (no cycles), edges not lengths or vertices, with first 4 edges correct

7 edges

CD (or ED) 5th

All correct

M1

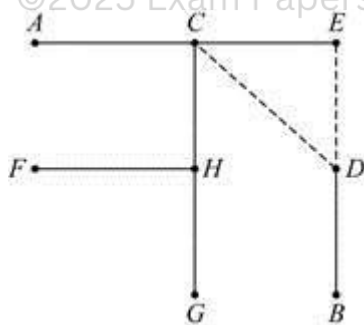
B1

A1

A1

4

(ii)



CD, ED either of these lines

ST with 5+ edges, connected, no cycles

Correct, including labelling

M1

A1

2

(iii) 75(p)

B1

1

- (b) Delete CH, HG, HF and add FA and one of GC, GA, GD, GF

Deleting their edges connected to H , and adding edges to make a ST with 6 edges

M1

or
a ST with 6 edges not including H
(either as a list or a diagram)

70(p)

Note: 70 scores 2/2

A1

2

[9]

Q6.

- (a) eg $A \ B \ C \ D \ E \ F \ A$

*Any tour ABA or better, any start vertex but not revisiting a vertex
May be shown in a labelled diagram of a cycle (eg triangle ABC)*

M1

*With all vertices visited
May be shown in a labelled diagram of a cycle*

A1

2

- (b) (i) $F \quad D \quad C \quad A \quad B \quad E \quad F$
 (20) (15) (5) (25) (15) (15)
 (= 95)

AG

Any tour, start/finish at F

M1

Visits all vertices

m1

Correct order

A1

3

If solution shown solely on matrix, then order of selection of vertices must be shown

- (ii) Tour

*"It's an answer", "a cycle", "it works"
"it's possible ..."*

E1

May be improved on

“Can’t be worse”, “not necessarily best”,

“could be improved”

Not “can be improved”

E1

2

(c) $F \quad E \quad C \quad A \quad B \quad D \quad F$

Tour $FE(ABCD \text{ in any order with } B \text{ before } D)F$

M1

(30) (7) (5) (25) (11) (10)

Correct order

= 88

A1

B1

3

If solution shown solely on matrix, order of selection of vertices must be shown

[10]

Q7.

(a) Odds A, B, C, D

PI (but A, B, C, D must be mentioned)

M1

Considering 3 sets of pairings of odd vertices, eg AB with CD etc

m1

$$\left. \begin{array}{l} AB + CD = 270 + 270 = 540 \\ AC + BD = 290 + 290 = 580 \\ AD + BC = 260 + 270 = 530 \end{array} \right\}$$

A1 for 2 correct, A2 for all correct

A2,1,0

Repeat AD, BC

Follow through their shortest pairing

PI by adding 530 to 1920

*Or AEHD or DHEA **and** BFGC or CGFB*

listed in any route

A1F

(Length = $1920 + 530 =$) 2450 (metres)

B1

6

(b) Repeats *BC*

PI by BFGC or CGFB listed in a complete route or adding 270 / subtracting 260

E1

(Length = $1920 + 270 =$) 2190 (metres)

$2450 - 260 = 2190$

(2190 with no evidence scores E0B1)

B1

2

(c) (i) Min. repeat *AD*

PI by AEHD or DHEA listed in a complete route or adding 260 / subtracting 270

E1

(Length = $1920 + 260 =$) 2180 (metres)

$2450 - 270 = 2180$

(2180 with no evidence scores E0B1)

B1

2

(ii) *B, C*

Condone start at B, finish at C (or reverse)

B1

1

[11]

Q8.

(a) *A, C, D, F* odd nodes

May be implied

B1

$$AC + DF = 18 + 22 = 40$$

M1

$$AD + CF = 32 + 30 = 62$$

A2,1,0

$$AF + CD = 12 + 30 = 42$$

Repeat $AC + DF$

May be implied

B1

$$\text{Total } 164 + 40 = 204$$

B1

6

(b) Start / finish A/C

$\therefore$ Repeat DF

Or subtract AC

B1

$$\text{Total } 164 + 22 = 186$$

B1

2

(c) (i) Shortest pair AF

B1

$$\text{Distance} = 164 + 12 = 176$$

B1

2

(ii) Start / Finish at C/D

May be listed in a route

B1

1

[11]

Q9.

- (a)
- | | |
|------|-------|
| AB | or 20 |
| AC | 25 |
| BD | 30 |
| CE | 35 |
| EF | 40 |
| FI | 35 |
| HI | 30 |
| IK | 35 |
| HG | 40 |
| HJ | 45 |

SCA

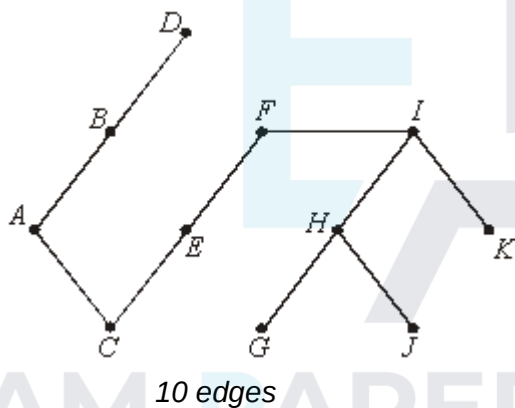
M1

10 edges

			B1	
	<i>BD third</i>			
			A1	
	<i>CE fourth</i>			
			A1	
	<i>All correct</i>			
			A1	5

(b)	335		B1	1
-----	-----	--	----	---

(c)



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(d)	Add AE	+40		
	<i>Adding AE, deleting CE/CA</i>			
			M1	
	Delete CE	−35		
	Extra	+5		
	<i>15 with no working (M1, A0)</i>			

A1	2
----	---

[11]

Q10.

(a)	(i)	$(ABDK) = 3$		
			B1	1
	(ii)	$MST = (n - 1)$		

M1

$$= 15$$

A1

2

- (b) (i) Odd vertices A and P

Route length = all edges + repeated edges (answer > 30)

M1 M1

Route length = all + (3)

A1

$$(33 + 3) = 36$$

B1

4

- (ii) Repeated: *AE, EL, LP*

B1

1

- (iii) *E* and *L* now odd

M1

$$\text{Their}(33) - 2 + 1 = 32$$

A1F

2

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[10]

Q11.

- (a) (i) RISC RWICWCR
or equiv.

M1

2

Must have RCW reference

A1

- (ii) 1000

B1

1

- (b) (i) RW makes odd

B1

2

vertices even

E1

- (ii) $I < 80 + 80$
inequality (implied)

M1

2

$$I = 159$$

SC/ 160 B

A1

[7]

Q12.

- (a) PQ, PT 30, 30

M1

TU 30

A1

ST 35

A1

QR 40
 UV 50

A1

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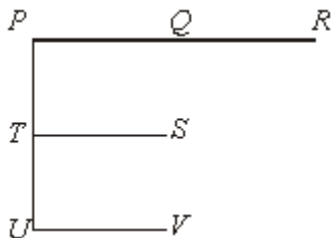
Total cost 215p

For cao: B2

B1

5

- (b)



M1 A1ft

2

- (c) PQ allowed but QR not

M1

Cheapest alternative to reach R is by SR
 raising cost by 5p to 220p

A1ft

2

[9]

Q13.

(a) AF 10

M1 A1

EF 10

DE 15

AB 20

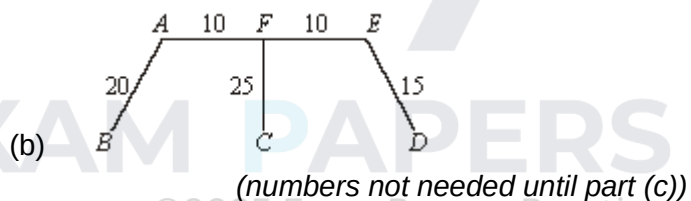
CF 25

A1

A1

A1

5



M1 A1

2

(c) All points are linked by edges in the spanning tree with maximal paths

M1

$20 + 10 + 25$, $20 + 10 + 10 + 15$ and

A1

$25 + 10 + 15$ all less than 60 minutes

A1

3

[10]

Q14.

(a) (i) Six odd vertices $PRSTUW$
 Just B1 for "odd vertices"

B1

Pairing them off take at least 3 tracks

B1

2

- (ii) Want to pair off PRSTUW to include PS
(or use any sensible short-cuts)

M!

or PU:

$PS \ RT \ UW \quad PU \ RS \ TW$

A1

$PS \ RU \ TW \quad PU \ RT \ SW$

$PS \ RW \ TU \quad PU \ RW \ ST$

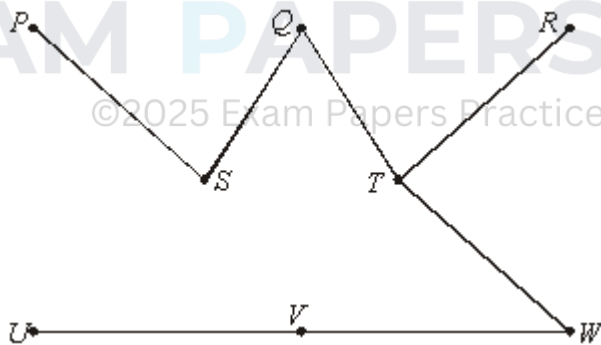
A1

Repeating $PS \ RT \ UV/VW$ takes $4 + 4 + 4$ and is clearly unbeatable as each joined pair adds at least 4.

A1

5

- (b) $UV \ VW \ QS \ PS \ QT \ RT \ WT$: 23



M1 A1

M1 A1

4

- (c) (i) Trainspotter's cycle length $\geq$
 $25 + 25 + \text{minimum connector length}$
 $= 50 + 23 = 73$

M1

- (ii) 73 not possible because (unique) minimum connector is not a path.

A1

Switching QT to QR makes it a path 1 mile longer, giving shortest round route:

B1 M1

e.g. Home $PSQRTWVU$ Home

A1

5

[16]

Q15.

(a) AC 17 AE 18

AD 19 BC 20

BF 21

Total length 95 miles

M1/A1

A1

A1

B1

5

(b) AC , AE , AD as before

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M1

Then AF (22) and BF , increasing total by

2

A1

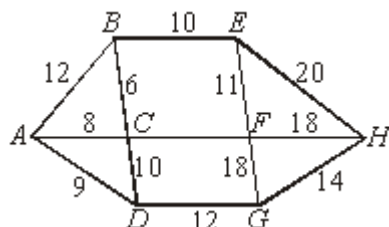
A1

3

[8]

Q16.

(a)



M1 A1

2

(b)

M1

BC, AC, AD,
lowest edges

A1

BE, CF (or vice-versa),
Selected 10s

A1

DG and GH
Rest

A1

Total cost = $6 + 8 + 9 + 10 + 10 + 12 + 14 = \text{£}69$

B1

5

(c) (i) The four numbers in C's row should be halved

B1

(ii) A sends information: lowest *AC as before*

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M1ft

Use *CB, CF* as before

A1

Use *CD* instead of *AD*

A1

New lowest cost = $3 + 8 + 5 + 10 + 5 + 12 + 14 = \text{£}57$

B1

5

[12]

Q17.

(a) *AD* 5

Start

M1

2

AF 6

A1

FG 6

A1

AE 7

Dealing with the 7s

M1

2

EC, EH, BC (all 7) in some appropriate order

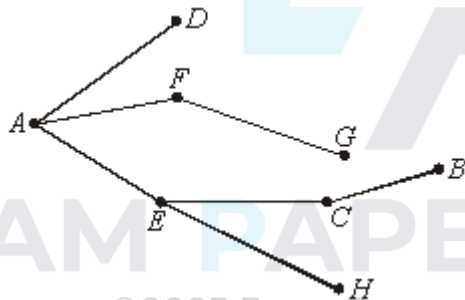
Total 45

Or matrix method

M1

2

(b)



M1 A1

$$AECB = 7 + 7 + 21$$

B1ft

3

(c) A to B too long

B1

Need to grit BD instead of BC

M1

(adding just one extra mile to the gritting)

A1

3

[12]

Q18.

- (a) **AE, AG, AC, AF, BE, BD**
 $\longleftrightarrow \quad \longleftrightarrow$

M1

method implied by order

A1

$$= 10 + 10 + 20 + 20 + 60 + 45 = \text{£}1.65$$

M1A1B1

5

- (b) First not involving **A** is **BE**, beaten by **AB**: rest same
M1 only for simply halving A's values in (a)

M1A1

$$\therefore \text{cost} = 5 + 5 + 10 + 10 + 40 + 45 = \text{£}1.15$$

M1A1

4

[9]

Q19.

- (a) Odd vertices – A, C, D, G

M1

Repeat ABEC and DFG
 can be implied

A2

$$2300 + 300 + 200 = 2800$$

A1

- (b) $2 \times 2300 = 4600$ – all even nodes so transversal

M1 A1

Explanation

B1

- (c) Digraph needed, but still transversal

B1

Route or argument such as “ins” = “outs” at each vertex

B1

[9]

Examiner reports

Q3.

Part (a) was well-answered – students had been well-drilled in the rote use of an elementary algorithm, although arithmetical accuracy was sometimes weak. However, in the remainder of the question, students were required to do a little elementary analysis and thinking and many failed completely. It appears that these modified situations are not handled well. Students need to spend time studying semi-Eulerian situations.

Q4.

Generally this was a very high scoring question with many full marks gained. Even candidates who earned very few marks elsewhere on the paper usually managed a high mark.

In part (a) it was pleasing that the answer to part (a)(i) was more clearly presented, from almost all candidates, than a similar answer on last summer's paper. Very few forgot to label their spanning tree in part (a)(iii).

Part (b) presented no problem at all. Most subtracted two lengths and added an acceptable replacement - really the only error in this part was to add the wrong length.

A considerable minority reworked the question from scratch with B removed.

Q5.

This question was a problem for some candidates, with their failure to understand Prim's or Kruskal's algorithms and their persistence in finding a route. However, for the better candidates, the question was generally well answered. It was apparent that candidates had attempted a correct application of Prim's algorithm but totally failed to indicate the **edges** involved correctly. These candidates had often done all their working within the table of the question. Candidates should be aware that even where they have attempted to show the order of choice of vertices within the table, it is not always clear to examiners precisely which edges are involved. Whatever method of working, there should be an explicit statement of the order of choice of vertices.

In part (b), many candidates deleted only two edges from their minimum spanning tree instead of the required three. Others failed totally to make clear which edges were being deleted and added, offering only a sequence of apparently random numbers which was futile given the number of repeat values in the table.

Q6.

This work was well known, continuing the improvement from previous series. The weakest candidates often failed to complete tours. There were very few answers presented in matrix form – a considerable proportion of these failed to provide indication of order. In part (a), a surprising number of candidates omitted to give an example, opting to give a general description or definition of a tour. Part (b)(i) was usually correct. In part (b)(ii), as with similar past questions, there was a majority view that the current figure *can* be improved rather than *might* be. Some still failed to pick up the 'tour' mark. There were many correct responses to part (c) of the question.

Q7.

In part (a), the method was well known and most candidates scored heavily, although a surprising number failed to get the correct total for all three pairs.

After that there were a lot of problems and the question discriminated well. Roughly a third of candidates thought that parts (b) and (c)(i) could be solved by inspection and about half of them were successful. Another third resorted to attempting to work out the route, apparently by trial and error, and were usually unsuccessful. The remainder either omitted the questions or wrote a random number down. The clear majority answer for part (c)(ii) was *A, B, C* and *D*. Once again, the able candidates were usually correct.

Q8.

This question proved to be a good discriminator. In part (a) candidates realised that the method involved pairing odd vertices, however many candidates failed to find the minimum pairing connecting the odd vertices. Candidates must realise that they must consider the shortest distances connecting vertices when solving a Chinese postman problem. Parts (b) and (c) were poorly answered. Many candidates produced what appeared to be a random selection of numbers, with little thought as to the requirements of the question. These responses showed a lack of an in-depth understanding of the theory relating to Chinese postman problems.

Q9.

Many candidates failed to indicate that Prim's algorithm had been used. Candidates must list the order in which edges are added to their spanning tree. Also a significant number of candidates started using the correct algorithm but failed to persevere to the end of the question with a loss of marks ensuing.

Q10.

Part (a) was well answered.

Part (b) proved to be more challenging. Candidates were able to make a good attempt at part (i), although a significant number used $16 + 2$ or 3 . In part (ii), a common mistake was to omit the edge *EL*. Part (iii) proved to be a good discriminator. Candidates failed to realise that by removing *AE* and *LP*, then *E* and *L* would become the odd vertices.

Q11.

The question was well answered although a number of candidates did not realise that the quickest route from *R* to *W* was via *C*. Some candidates did not read part (b)(ii) carefully as they failed to give their answer as an integer value.

Q13.

This was extremely well done.

Q16.

All candidates could construct the network and they seemed to find the pre-printed vertices helpful. The vast majority could also construct a minimum connector, although a sizeable minority used Prim's algorithm rather than Kruskal's. The specification states that candidates should know both, and, when given a network, Kruskal's is marginally easier than Prim's. In adapting their answers for part (c) too many candidates simply adapted the costs in the same tree that they had constructed in (b) without realising that one edge

needed changing too (CD now being cheaper than AD).

Q18.

The application of Prim's algorithm is usually found to be very straightforward, but the matrix version upset many candidates. Since "or otherwise" was allowed, the candidates could have used a network if they preferred. Even those candidates who got part (a) right failed to realise that the changes in part (b) make it an essentially different problem: simply halving some of the links in the answer to part (a) is not good enough.

Q19.

Most candidates attempted this question, which discriminated well.

In part (a), a surprisingly large proportion of candidates did not quote the name (or provide any evidence) of the "appropriate algorithm" required (route inspection). Many did not appear to have considered odd vertices at all; there was little evidence of computing distances to obtain the optimal pairing. Some candidates noted that AB must start and BA end the route. They eliminated the arc AB , making B odd. This was a perfectly valid simplification.

In part (b) it was intended that candidates would realise that the nodes were all even, so that the route was traversable and twice the original length. Some candidates produced a correct route and corresponding correct distance, added up arc by arc. Demonstrated calculation of the distance was accepted as a valid justification for the answer, although it was far from elegant.

Many candidates realised that the distance in part (c) was the same as in part (b). Some were able to justify this conclusion, obtaining the final mark, by giving their route correctly. There were some attempts at the (difficult) justification of the result in general.

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